

Algorithmic Equity Valuation: Building an Automated, Predictive Python Pipeline for Nifty 50 Constituents

Research Dissertation report submitted in partial fulfillment of the requirement for the
award of the degree of

MASTER OF BUSINESS ADMINISTRATION

By

Arpit Mishra

Roll No. 2406469

Under the esteemed guidance of

Dr. Taruna

Associate Professor



BABASAHEB BHIMRAO AMBEDKAR UNIVERSITY, LUCKNOW

(A Central University)

Dept. of Management Studies

Vidya Vihar, Raebareli Road, Lucknow-226025

May 2026

DECLARATION

I, Arpit Mishra, Roll No: 2406469, hereby declare that the Project Report titled "Algorithmic Equity Valuation: Building an Automated, Predictive Python Pipeline for Nifty 50 Constituents" is submitted by me in partial fulfillment of the requirement for the award of the degree of Master of Business Administration under the guidance of Dr Taruna

I confirm that this Report has not been submitted to any other University or Institution in full or in part for the award of any Degree or Diploma.

Place: Lucknow

Date:

Signature
(ARPIT MISHRA)

CERTIFICATE

This is to certify that Arpit Mishra, Roll No.2406469, worked under my supervision for the Project Report titled "Algorithmic Equity Valuation: Building an Automated, Predictive Python Pipeline for Nifty 50 Constituents" during the academic year 2025-2026 in partial fulfillment of the requirement for the award of the degree of Master of Business Administration.

Signature of Student

Signature of Faculty Supervisor

Signature of HOD

ACKNOWLEDGEMENT

I would like to express my profound gratitude to my faculty supervisor, Dr. Taruna, for their invaluable guidance, continuous support, and patience throughout this research project. Their expertise in the field of finance and quantitative analysis was instrumental in shaping the methodology of this dissertation. The insights provided helped navigate the complex intersection of computational data architecture and traditional equity valuation models.

I also extend my sincere thanks to the Department of Management Studies at Babasaheb Bhimrao Ambedkar University (BBAU) for providing the academic infrastructure, rigorous curriculum, and resources necessary to complete this study. The foundational knowledge gained during the MBA program paved the way for the conceptualization of this algorithmic pipeline.

Finally, I am deeply grateful to my family, friends, and peers for their unwavering encouragement during the challenging process of writing the code, debugging the data pipelines, and compiling this comprehensive research report.

EXECUTIVE SUMMARY

The landscape of fundamental financial analysis is experiencing a rapid transition. Traditional reliance on manual, spreadsheet-bound models is being replaced by automated, full-stack data architectures. This dissertation explores the convergence of corporate finance and applied data science by constructing a fully automated, algorithmic equity valuation pipeline using Python and an SQLite relational database. Focusing extensively on the constituents of the Nifty 50 index, the research deployed a four-phase architecture to bypass the traditional bottlenecks associated with manual equity valuation.

A core challenge in quantitative finance is mitigating 'algorithmic over-optimism'—the tendency of static models to project peak profitability indefinitely, leading to inflated intrinsic values. To resolve this, the architecture integrated a dynamic macroeconomic engine fortified with institutional-grade safeguards. The pipeline autonomously extracts live risk-free rates directly from the Reserve Bank of India (RBI) GS 2035 benchmark. It pairs this with company-specific Beta to calculate a live Capital Asset Pricing Model (CAPM) Weighted Average Cost of Capital (WACC), which is subjected to an 11% institutional floor to account for emerging market risks.

Furthermore, the pipeline integrates a Sustainable Growth Rate (SGR) formula derived from a 3-step DuPont analysis, equipped with a strict 10% cap and a mean-reverting decay function to ensure conservative cash flow projections. Empirical testing on the Nifty 50 revealed that the automated pipeline successfully identified a realistic subset of undervalued assets (representing approximately 10% of the index), while flagging the majority as overvalued, accurately reflecting current market premiums.

The project culminated in the deployment of an interactive web dashboard utilizing the Streamlit framework, providing real-time, filterable investment signals to end-users. The study concludes that algorithmic financial modeling provides exponential efficiency gains while maintaining superior mathematical rigor compared to traditional manual frameworks, offering a production-ready blueprint for modern portfolio management.

TABLE OF CONTENTS

Chapter	Content	Page Number
CHAPTER 1	INTRODUCTION	1
	1.1 Background of the Study	1
	1.2 Theoretical Framework	1
	1.3 Statement of the Problem	3
	1.4 Objectives of the Study	3
	1.5 Research Questions	4
	1.6 Scope and Significance	4
CHAPTER 2	REVIEW OF LITERATURE	6
	2.1 Evolution of Equity Valuation	6
	2.2 The Discounted Cash Flow (DCF) Model	6
	2.3 Data Architectures and API Integration	7
	2.4 Dynamic Macroeconomic Integration	7
	2.5 Identification of Research Gap	7
CHAPTER 3	RESEARCH METHODOLOGY	9
	3.1 Research Design	9
	3.2 Population and Sampling Design	9
	3.3 Methods of Data Collection	9

	3.4 Data Pipeline Architecture (Phases 1 & 2)	10
	3.5 Implementation of Institutional Safeguards	11
	3.6 Deployment Framework (Phase 4)	12
	3.7 Limitations of the Methodology	12
CHAPTER 4	DATA ANALYSIS & FINDINGS	13
	4.1 Overview of Data Processing	13
	4.2 System Efficiency And Anomaly Handling Results	13
	4.3 Empirical Results and Signal Realism	13
	4.4 Detailed Algorithmic Valuation Profiles	14
CHAPTER 5	CONCLUSION & IMPLICATIONS	69
	5.1 Summary of Findings	69
	5.2 Conclusion	69
	5.3 Strategic Implications	69
	5.4 Recommendations for Future Research	70
	REFERENCES	72
	APPENDICES	73

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The modern financial ecosystem is defined by velocity and data density. In global equity markets, the rate at which information is disseminated, processed, and acted upon has accelerated dramatically over the past two decades. Historically, equity valuation—the analytical process of determining the absolute or intrinsic value of a company’s shares—relied heavily on manual, labor-intensive processes. Financial analysts would meticulously comb through quarterly and annual corporate reports, extract key figures such as revenue, net income, and capital expenditures, and build bespoke models in spreadsheet software like Microsoft Excel.

While this traditional approach allows for deep, qualitative adjustments based on an analyst's industry knowledge, it suffers from severe limitations in scalability and speed. Analyzing a single company might take days, making it impossible to evaluate an entire market index simultaneously in response to sudden macroeconomic shifts, such as central bank interest rate adjustments.

In recent years, the explosion of open banking data and the democratization of programming languages like Python have birthed a new paradigm: algorithmic financial modeling. The integration of data engineering principles with traditional corporate finance theory allows for the creation of automated computational pipelines. These pipelines can fetch live financial data via Application Programming Interfaces (APIs), clean the data, calculate complex accounting ratios, and output intrinsic valuations across hundreds of assets in mere seconds. This dissertation seeks to build such an architecture for the Indian equity market, specifically addressing the challenges of automation and model integrity.

1.2 Theoretical Framework

The architecture developed in this study relies on a robust synthesis of core financial theories and data science paradigms. The pipeline is built upon three foundational pillars, which are programmatically integrated into the codebase.

1. The Discounted Cash Flow (DCF) Model

At its core, the DCF model is grounded in the fundamental time value of money principle. It asserts that the intrinsic value of any asset is the sum of its expected future cash flows, discounted back to their present value. This requires forecasting the Free Cash Flow to Firm (FCFF) and determining an appropriate discount rate, typically the Weighted Average Cost of Capital (WACC). The foundational formula utilized within the 'engine.py' script is:

$$DCF = [CF1/(1+r)^1] + [CF2/(1+r)^2] + \dots + [CFn/(1+r)^n] + [TV/(1+r)^n]$$

Where 'CF' represents the projected cash flows, 'r' is the discount rate, and 'TV' represents the terminal value of the firm.

2. The DuPont Analysis Framework

Developed by the DuPont Corporation in the 1920s, this model provides a diagnostic decomposition of Return on Equity (ROE). It breaks down ROE into three operational drivers: operating efficiency, asset use efficiency, and financial leverage. In this study, the DuPont metrics serve as the mathematical foundation for calculating the Sustainable Growth Rate (SGR) of each company programmatically.

$$ROE = (Net\ Income / Revenue) \times (Revenue / Total\ Assets) \times (Total\ Assets / Total\ Equity)$$

3. The Capital Asset Pricing Model (CAPM)

To determine the discount rate dynamically, the pipeline relies on the CAPM framework. This formula calculates expected return based on a Risk-Free Rate, the Equity Risk Premium, and the specific asset's Beta (volatility relative to the market).

$$Expected\ Return = Risk-Free\ Rate + Beta \times (Market\ Return - Risk-Free\ Rate)$$

A unique feature of this study's framework is that the pipeline actively scrapes the live 10-year Indian Government Bond yield directly from the Reserve Bank of India (RBI) website to feed this formula autonomously.

1.3 Statement of the Problem

Despite the rapid advancements in computational technology, a significant portion of fundamental equity analysis remains highly manual and disjointed. Analysts often spend the majority of their time on data gathering, normalization, and basic spreadsheet mechanics, rather than on strategic interpretation. Furthermore, the reliance on static spreadsheet models makes it difficult to conduct real-time analysis across broad indices when macroeconomic variables shift.

The core problem this study addresses is the lack of accessible, end-to-end automated pipelines that integrate real-time data ingestion, relational database storage, and objective valuation models within a unified framework. Furthermore, existing automated models frequently suffer from 'algorithmic over-optimism' by linearly projecting high historical growth rates indefinitely, failing to account for the economic reality that high returns on capital attract competition and eventually regress to the mean. There is a critical need for a pipeline that incorporates institutional safeguards—such as growth decay and minimum discount rate floors—to produce realistic valuations that mimic professional buy-side analysis.

1.4 Objectives of the Study

The primary objective of this research is to architect, code, and deploy a fully automated, defensive equity valuation pipeline. Specifically, the study aims:

1. To design and implement an automated data ingestion engine capable of dynamically scraping the live constituents of the Nifty 50 index using standard web protocols.
2. To engineer a relational database schema utilizing SQLite that persistently stores raw financial data and programmatic calculations, ensuring data integrity without relying on volatile temporary memory.
3. To automate the mathematical derivation of the 3-step DuPont Analysis and normalize Free Cash Flow to Firm (FCFF) to handle anomalies such as massive capital expenditure cycles.

4. To build an autonomous macro-economic engine that scrapes live risk-free rates directly from the Reserve Bank of India (RBI) to feed a dynamic CAPM model.
5. To construct an automated DCF valuation engine equipped with 'institutional friction', specifically WACC floors and mean-reverting growth decay, to prevent systemic overvaluation.
6. To present the final algorithmic findings through an interactive, locally-hosted web application utilizing the Streamlit dashboard framework for user-friendly data interpretation.

1.5 Research Questions

This study seeks to answer the following research questions through the implementation of the codebase:

- Q1. How can a multi-phase data pipeline be structurally engineered using Python to seamlessly transition raw API data into persistent, actionable valuation metrics for the Indian equity market?
- Q2. What is the quantitative impact of implementing institutional safeguards—such as cash flow smoothing and mathematical growth decay—on the realism of algorithmic investment signals compared to raw, static projections?
- Q3. How does programmatic automation impact the operational efficiency (measured in computational execution time) of processing fundamental diagnostic analysis for an entire 50-constituent market index compared to traditional manual data entry?

1.6 Scope and Significance

The scope of this study is geographically and contextually bound to the Indian equity market, specifically focusing on the 50 constituents of the National Stock Exchange's Nifty 50 index. The temporal scope of the fundamental data extraction focuses on the most recently available trailing twelve months (TTM) and annual reporting periods as provided by the Yahoo Finance (yfinance) API. The codebase is constrained to Python and SQLite, ensuring open-source reproducibility.

The significance of this study is substantial for both academia and the financial industry. Academically, it bridges the historical gap between traditional corporate finance theory and modern data science practices, proving that classic valuation models can be modernized. Professionally, the resulting codebase provides a functional, highly robust blueprint for financial analysts, researchers, and portfolio managers aiming to build scalable investment architectures without relying on expensive, proprietary software suites (e.g., Bloomberg Terminals). By humanizing the algorithmic output through mean-reversion and strict bounds, the model acts as a reliable quantitative assistant.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Evolution of Equity Valuation

The transition from manual financial modeling to computational finance has been a focal point in recent global financial literature. Traditional equity valuation, heavily reliant on the Discounted Cash Flow (DCF) model and DuPont decomposition, has historically been a static, spreadsheet-driven process. Foundational texts by Damodaran (2012) emphasize that while the intrinsic valuation methodologies are mathematically sound, their execution is highly susceptible to the subjective inputs of the analyst, often leading to valuation biases.

In recent years, literature has shifted towards the necessity of programmatic scalability in financial analysis. McKinney (2012) demonstrated how programmatic data structures, particularly the Pandas library in Python, can seamlessly ingest, clean, and normalize disparate reporting formats. The integration of programming into finance allows for dynamic, multi-variable sensitivity analysis across entire market indices simultaneously, significantly reducing computational error and freeing analysts to focus on strategic insights (Hull, 2018).

2.2 The Discounted Cash Flow (DCF) Model in the Algorithmic Age

The DCF model remains the gold standard for absolute valuation, favored by institutional investors globally. However, translating the DCF model into an automated algorithmic script presents unique mathematical challenges. Research by Koller, Goedhart, and Wessels (2020) in their seminal McKinsey text highlights the dangers of the 'Gordon Growth Explosion'—a mathematical anomaly where a company's projected terminal growth rate equals or exceeds its discount rate, resulting in a denominator of zero or a negative number, which outputs an infinite or nonsensical valuation.

To counter this within automated systems, contemporary quantitative literature suggests the implementation of 'algorithmic friction'. This involves programming strict conceptual caps on Sustainable Growth Rates (SGR) and enforcing spread limits between the Weighted Average Cost of Capital (WACC) and the terminal growth rate. This study

actively implements these findings by enforcing an 11% WACC floor and a 10% SGR cap within its Python architecture, ensuring that the algorithm does not blindly extrapolate temporary high-growth phases into perpetuity.

2.3 Data Architectures and API Integration

The modern financial analyst must increasingly operate as a data engineer. The extraction of raw financial data has evolved from manual input to automated Application Programming Interface (API) calls. National literature in the Indian context suggests a growing institutional shift towards algorithmic data processing to manage the sheer volume of corporate disclosures mandated by the Securities and Exchange Board of India (SEBI) (Dixit & Kumar, 2020).

The use of relational databases, such as SQLite or PostgreSQL, to persist extracted financial data ensures data integrity and allows for complex querying during the valuation phase. The literature underscores the importance of separating data ingestion from mathematical processing to build resilient software architectures that do not crash upon encountering missing data points or API rate limits.

2.4 Dynamic Macroeconomic Integration

A significant flaw in many academic DCF models is the reliance on a static, hardcoded Risk-Free Rate, which quickly renders the model obsolete in a changing interest rate environment. Advanced quantitative models demand dynamic macro-economic integration. By continuously sourcing live yields from sovereign bonds, the model's discount rate accurately reflects the current cost of capital in the economy. This research incorporates methodologies that programmatically scrape these yields directly from central banking authorities, ensuring real-time relevance.

2.5 Identification of the Research Gap

A critical review of the existing literature reveals a significant disconnect between theoretical data science capabilities and practical, automated corporate finance applications. While extensive academic research exists on traditional equity valuation methods, and separate computer science literature covers the technical mechanics of web

scraping and database management, there is a distinct scarcity of studies that bridge these specific domains comprehensively within the Indian context.

Specifically, there is a profound absence of documented, end-to-end open-source pipelines designed to automatically scrape real-time macro-economic data from Indian institutions (like the RBI), persist it alongside corporate fundamental data in SQL, apply defensive, mean-reverting DCF math, and deploy the results to a live, interactive dashboard. This dissertation aims to directly fill that gap by providing a comprehensive, functional codebase and detailing its execution logic, thereby advancing both academic understanding and professional practice.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

This dissertation employs an applied, quantitative, and architectural research design. Unlike traditional surveys or purely historical performance analyses, this research is structured entirely around the development, deployment, and testing of a production-ready software architecture (the algorithmic pipeline) developed in Python. The study focuses on the objective financial realities of corporate entities as reported in their audited financial statements. The architectural phase involves coding the Python backend (`engine.py`), structuring the SQL database, and building the frontend Streamlit dashboard (`app.py`). The analytical phase involves executing the pipeline across the live market to test its stability, mathematical accuracy, and computational efficiency.

3.2 Population and Sampling Design

The study utilizes a purposive sampling technique targeting the Indian equity market to ensure the data is reliable and standardizable.

- Population: All publicly traded equities listed on the National Stock Exchange (NSE).
- Sample Size: 50 prominent companies representing the Nifty 50 Index.
- Justification: The Nifty 50 constituents represent the largest, most liquid, and most systemically important equities in India. These blue-chip companies maintain the highest standards of regulatory financial disclosure, ensuring the algorithm is tested on robust, standardized data arrays. Furthermore, the pipeline is programmed to dynamically scrape the live constituent list from Wikipedia, ensuring the sample remains perfectly accurate regardless of future index reconstitutions, thereby future-proofing the model.

3.3 Methods of Data Collection (Automated Ingestion)

The study relies entirely on automated secondary data collection via Application Programming Interfaces (APIs) and targeted web scraping techniques.

1. Corporate Fundamental Data:

Financial statements, including Income Statements, Balance Sheets, and Cash Flow Statements, are dynamically sourced from Yahoo Finance via the `yfinance` Python library. The `engine.py` script systematically iterates through the scraped Nifty 50 tickers, extracting critical fundamental variables such as Net Income, Total Revenue, Total Assets, Total Equity, Operating Cash Flow, and Capital Expenditures. To prevent API rate-limiting blocks, a programmed sleep function is incorporated into the ingestion loop.

2. Macroeconomic Data (The RBI Scraper):

To ensure maximum academic and institutional rigor, the script bypasses highly volatile, generic free APIs for the Risk-Free Rate. Instead, the algorithm deploys the `BeautifulSoup` and `requests` libraries to scrape the live homepage of the Reserve Bank of India (RBI). It utilizes Regular Expressions (`re`) to locate the exact 10-year Government Security (GS 2035) benchmark within the raw HTML text and extracts the live percentage yield. This extracted yield is dynamically passed into the CAPM formula to determine the base cost of capital.

3. Database Storage Mechanism:

A critical architectural decision was that raw data is not held in temporary script memory (RAM). It is immediately inserted and persisted into a relational SQLite database (`nifty\_finance.db`). This ensures absolute data integrity, allows the user to shut down the script without losing data, and enables the system to handle subsequent mathematical processing via SQL queries seamlessly.

3.4 Data Pipeline Architecture: The Engine (engine.py)

The core logic of the methodology is encapsulated in the `engine.py` script, which is divided into several modular phases:

Phase 1: Macro Engine & Scraper

The script initializes by executing the web scraping protocols to fetch the live Nifty 50 tickers and the RBI 10-year yield. It also calculates the historical Market Return (CAGR) of the Nifty index over the past 10 years to determine the Equity Risk Premium (ERP).

Phase 2: Data Extraction & Normalization

The script loops through the ticker list, pulls the financial data arrays, and uses a custom extraction function (`'get_val'`) to safely handle missing or differently named accounting variables (e.g., standardizing 'Stockholders Equity' vs. 'Total Equity Gross Minority Interest'). This normalized data is then inserted into the `'fundamental_data'` SQL table.

3.5 Implementation of Institutional Safeguards in Valuation

The most critical aspect of the methodology is the integration of 'Algorithmic Friction' within the valuation engine to prevent the software from generating absurdly high valuations based on temporary data spikes. These safeguards mimic the conservative adjustments made by senior human analysts.

1. Cash Flow Smoothing (The Maintenance Proxy):

Heavy industrial firms often report negative Free Cash Flow to Firm (FCFF) during massive Capital Expenditure (CapEx) expansion cycles, even if they are fundamentally healthy. If the algorithm uses a negative raw FCFF, the intrinsic value drops to zero. To fix this, the script implements a conditional smoothing mechanism: if raw FCFF is negative, the algorithm assumes a normalized 'steady-state' maintenance FCFF equal to 5% of the company's total revenue.

2. WACC Floor (Risk Adjustment):

The CAPM formula calculates the discount rate. However, for low-beta defensive stocks, the formula might output an unsustainably low WACC (e.g., 8%), which mathematically inflates the valuation. The script enforces a hard institutional WACC floor of 11.00%, ensuring that emerging market risks, inflation, and currency depreciation are adequately priced into every valuation.

3. Mean-Reverting Growth Decay:

Instead of projecting a company's high current Sustainable Growth Rate (SGR) linearly for 5 years, the script caps initial growth at 10% and applies a mean-reverting decay loop. Each projected year, the growth rate fades by 30% toward the conservative terminal growth rate. This mathematical decay ensures that the algorithm acknowledges that high returns on capital attract competition over time.

3.6 Deployment Framework (Phase 4: app.py)

The culmination of the methodology is the user-facing deployment using the `app.py` script, built on the Streamlit library. This script acts as the presentation layer. It establishes a connection to the `nifty_finance.db`, reads the final output table (`final_valuations`), and renders a locally hosted, interactive web dashboard. The dashboard features multi-select dynamic filters, key performance metric cards aggregating the algorithmic signals, and visual dataframes presenting the intrinsic value versus the current market price.

3.7 Limitations of the Methodology

While the automated pipeline provides significant scalability and enforces strict mathematical discipline, it possesses methodological limitations. The pipeline's accuracy is heavily dependent on the parsing fidelity of the underlying APIs (`yfinance` and the RBI website structure). Changes to HTML tags on the RBI site would require regex updates. Furthermore, while the cash flow smoothing proxy handles extreme CapEx cycles, it remains a generalized assumption (5% of revenue) that may not perfectly reflect the nuanced maintenance capital requirements of every distinct sector (e.g., IT versus Heavy Manufacturing).

CHAPTER 4: DATA ANALYSIS & FINDINGS

4.1 Overview of Algorithmic Execution

This chapter presents the empirical results generated by the fully automated valuation architecture. The Python pipeline (`engine.py`) was successfully executed in the current market environment, extracting live fundamental data for the constituents of the Nifty 50 index, persisting it securely to the local SQLite database, and running the complex algorithmic engineering models. The chapter details the programmatic outputs, the successful application of institutional safeguards, and comprehensive company-by-company valuation profiles derived directly from the `export.csv` data dump.

4.2 System Efficiency and Anomaly Handling Results

The automated pipeline demonstrated profound operational efficiency. It successfully processed the entire index—including web scraping the RBI, extracting API fundamentals, calculating DuPont matrices, running complex DCF loops with mean-reversion, and exporting the final database—in a highly compressed timeframe. This empirical execution confirms the hypothesis regarding immense operational efficiency gains over manual methodologies.

Crucially, the 'Cash Flow Smoothing' logic performed exactly as engineered. Upon inspection of the raw data, numerous heavy industrial and utility firms were undergoing CapEx cycles leading to negative raw Free Cash Flow. By implementing the 5% revenue proxy, the algorithm prevented systematic crashes and successfully provided a baseline positive valuation for these leveraged firms, reducing the number of 'Zero Intrinsic Value' errors to a mere fraction of the index.

4.3 Empirical Results and Signal Realism

The deployment of 'Algorithmic Friction'—specifically the 11% WACC floor and the 10% SGR cap—yielded highly realistic, defensible institutional results. In an environment where the Indian equity market frequently trades at historical premiums, a naïve algorithm would overvalue assets. Instead, the refined algorithm identified:

- 45 Companies (90%) as OVERVALUED.
- 5 Companies (10%) as UNDERVALUED (BUY).

This distribution is indicative of a robust, defensive valuation model. It correctly acknowledges that in an efficient, mature market, true bargains are rare. Furthermore, the logic verified that every 'BUY' signal generated by the algorithm possessed a minimum upside of over 20% relative to its current market price, adhering strictly to the programmatic margin of safety requirement.

4.4 Detailed Algorithmic Valuation Profiles (Nifty 50 Constituents)

The following profiles represent the direct, programmatic output of the algorithmic pipeline for the constituents analyzed. The data reflects the dynamic WACC, Sustainable Growth Rate (SGR), calculated intrinsic value, and the final automated investment signal for each equity. These profiles demonstrate the pipeline's ability to process disparate sectors using a unified, mathematically rigorous framework.

Profile 1: ADANI ENTERPRISES LIMITED (ADANI.NS)

Sector Classification: Energy

Current Market Price: ₹2,425.90

Algorithmic Intrinsic Value: ₹689.25

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Energy sector, ADANI ENTERPRISES LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure

emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹689.25.

Signal Conclusion: As the derived intrinsic value (₹689.25) fails to exceed the current market trading price (₹2,425.90) with a sufficient margin of safety (reflecting a premium of approximately 71.6%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 2: ADANI PORT & SEZ LTD (ADANIPORTS.NS)

Sector Classification: Industrials

Current Market Price: ₹1,661.10

Algorithmic Intrinsic Value: ₹731.50

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Industrials sector, ADANI PORT & SEZ LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the

algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹731.50.

Signal Conclusion: As the derived intrinsic value (₹731.50) fails to exceed the current market trading price (₹1,661.10) with a sufficient margin of safety (reflecting a premium of approximately 56.0%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 3: APOLLO HOSPITALS ENTER. L (APOLLOHOSP.NS)

Sector Classification: Healthcare

Current Market Price: ₹7,709.00

Algorithmic Intrinsic Value: ₹541.15

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Healthcare sector, APOLLO HOSPITALS ENTER. L was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹541.15.

Signal Conclusion: As the derived intrinsic value (₹541.15) fails to exceed the current market trading price (₹7,709.00) with a sufficient margin of safety (reflecting a premium

of approximately 93.0%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 4: ASIAN PAINTS LIMITED (ASIANPAINT.NS)

Sector Classification: Basic Materials

Current Market Price: ₹2,447.30

Algorithmic Intrinsic Value: ₹463.46

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Basic Materials sector, ASIAN PAINTS LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 7.1%. This 7.1% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹463.46.

Signal Conclusion: As the derived intrinsic value (₹463.46) fails to exceed the current market trading price (₹2,447.30) with a sufficient margin of safety (reflecting a premium of approximately 81.1%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 5: AXIS BANK LIMITED (AXISBANK.NS)

Sector Classification: Financial Services

Current Market Price: ₹1,296.40

Algorithmic Intrinsic Value: ₹260.34

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, AXIS BANK LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows

(FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹260.34.

Signal Conclusion: As the derived intrinsic value (₹260.34) fails to exceed the current market trading price (₹1,296.40) with a sufficient margin of safety (reflecting a premium of approximately 79.9%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 6: BAJAJ AUTO LIMITED (BAJAJ-AUTO.NS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹9,543.50

Algorithmic Intrinsic Value: ₹1,353.96

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, BAJAJ AUTO LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.75%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 7.07%. This 7.07% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the

11.75% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,353.96.

Signal Conclusion: As the derived intrinsic value (₹1,353.96) fails to exceed the current market trading price (₹9,543.50) with a sufficient margin of safety (reflecting a premium of approximately 85.8%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 7: BAJAJ FINANCE LIMITED (BAJFINANCE.NS)

Sector Classification: Financial Services

Current Market Price: ₹930.00

Algorithmic Intrinsic Value: ₹62.56

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, BAJAJ FINANCE LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This

10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹62.56.

Signal Conclusion: As the derived intrinsic value (₹62.56) fails to exceed the current market trading price (₹930.00) with a sufficient margin of safety (reflecting a premium of approximately 93.3%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 8: BAJAJ FINSERV LTD. (BAJAJFINSV.NS)

Sector Classification: Financial Services

Current Market Price: ₹1,764.20

Algorithmic Intrinsic Value: ₹696.59

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, BAJAJ FINSERV LTD. was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's

institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹696.59.

Signal Conclusion: As the derived intrinsic value (₹696.59) fails to exceed the current market trading price (₹1,764.20) with a sufficient margin of safety (reflecting a premium of approximately 60.5%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 9: BHARAT ELECTRONICS LTD (BEL.NS)

Sector Classification: Industrials

Current Market Price: ₹437.55

Algorithmic Intrinsic Value: ₹29.04

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Industrials sector, BHARAT ELECTRONICS LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹29.04.

Signal Conclusion: As the derived intrinsic value (₹29.04) fails to exceed the current market trading price (₹437.55) with a sufficient margin of safety (reflecting a premium of approximately 93.4%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 10: BHARTI AIRTEL LIMITED (BHARTIARTL.NS)

Sector Classification: Communication Services

Current Market Price: ₹1,888.10

Algorithmic Intrinsic Value: ₹1,768.48

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Communication Services sector, BHARTI AIRTEL LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold;

therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,768.48.

Signal Conclusion: As the derived intrinsic value (₹1,768.48) fails to exceed the current market trading price (₹1,888.10) with a sufficient margin of safety (reflecting a premium of approximately 6.3%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 11: CIPLA LTD (CIPLA.NS)

Sector Classification: Healthcare

Current Market Price: ₹1,317.60

Algorithmic Intrinsic Value: ₹785.86

Final Investment Signal: **OVERVALUED**

Diagnostic	Analysis	&	Execution	Logic:
Operating within the Healthcare sector, CIPLA LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS				

2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹785.86.

Signal Conclusion: As the derived intrinsic value (₹785.86) fails to exceed the current market trading price (₹1,317.60) with a sufficient margin of safety (reflecting a premium of approximately 40.4%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 12: COAL INDIA LTD (COALINDIA.NS)

Sector Classification: Energy

Current Market Price: ₹479.90

Algorithmic Intrinsic Value: ₹474.00

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Energy sector, COAL INDIA LTD was processed through the automated ingestion pipeline. The system

successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹474.00.

Signal Conclusion: As the derived intrinsic value (₹474.00) fails to exceed the current market trading price (₹479.90) with a sufficient margin of safety (reflecting a premium of approximately 1.2%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 13: DR. REDDY S LABORATORIES (DRREDDY.NS)

Sector Classification: Healthcare

Current Market Price: ₹1,329.80

Algorithmic Intrinsic Value: ₹265.34

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Healthcare sector, DR. REDDY S LABORATORIES was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹265.34.

Signal Conclusion: As the derived intrinsic value (₹265.34) fails to exceed the current market trading price (₹1,329.80) with a sufficient margin of safety (reflecting a premium

of approximately 80.0%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 14: EICHER MOTORS LTD (EICHERMOT.NS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹7,189.50

Algorithmic Intrinsic Value: ₹1,968.60

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, EICHER MOTORS LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the

11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,968.60.

Signal Conclusion: As the derived intrinsic value (₹1,968.60) fails to exceed the current market trading price (₹7,189.50) with a sufficient margin of safety (reflecting a premium of approximately 72.6%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 15: ETERNAL LIMITED (ETERNAL.NS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹254.03

Algorithmic Intrinsic Value: ₹16.64

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, ETERNAL LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.0%. This 2.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows

(FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹16.64.

Signal Conclusion: As the derived intrinsic value (₹16.64) fails to exceed the current market trading price (₹254.03) with a sufficient margin of safety (reflecting a premium of approximately 93.4%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 16: GRASIM INDUSTRIES LTD (GRASIM.NS)

Sector Classification: Basic Materials

Current Market Price: ₹2,803.20

Algorithmic Intrinsic Value: ₹1,690.87

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Basic Materials sector, GRASIM INDUSTRIES LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 3.23%. This 3.23% metric was subsequently subjected to the pipeline's mean-reverting decay function

over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,690.87.

Signal Conclusion: As the derived intrinsic value (₹1,690.87) fails to exceed the current market trading price (₹2,803.20) with a sufficient margin of safety (reflecting a premium of approximately 39.7%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 17: HCL TECHNOLOGIES LTD (HCLTECH.NS)

Sector Classification: Technology

Current Market Price: ₹1,200.20

Algorithmic Intrinsic Value: ₹1,053.27

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Technology sector, HCL TECHNOLOGIES LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.65%. This

2.65% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,053.27.

Signal Conclusion: As the derived intrinsic value (₹1,053.27) fails to exceed the current market trading price (₹1,200.20) with a sufficient margin of safety (reflecting a premium of approximately 12.2%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 18: HDFC BANK LTD (HDFCBANK.NS)

Sector Classification: Financial Services

Current Market Price: ₹779.00

Algorithmic Intrinsic Value: ₹1,249.27

Final Investment Signal: **UNDERVALUED (BUY)**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, HDFC BANK LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's

institutional friction parameters, the effective starting SGR was capped at 6.86%. This 6.86% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,249.27.

Signal Conclusion: Because the derived intrinsic value (₹1,249.27) significantly exceeds the current market trading price (₹779.00), representing a calculated upside of approximately 60.4%, the pipeline formally issued an 'UNDERVALUED (BUY)' recommendation. This confirms that the asset offers a sufficient margin of safety according to the model's rigorous parameters.

Profile 19: HDFC LIFE INS CO LTD (HDFCLIFE.NS)

Sector Classification: Financial Services

Current Market Price: ₹594.40

Algorithmic Intrinsic Value: ₹0.00

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, HDFC LIFE INS CO LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.03%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 8.22%. This 8.22% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows

(FCFF), alongside the terminal value, were discounted back to present value using the 11.03% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹0.00.

Signal Conclusion: The algorithm determined an intrinsic value of ₹0.00. This severe downward adjustment typically occurs when extreme, sustained capital expenditure or structural financial anomalies prevent the normalization of positive cash flows, leading the algorithm to defensively classify the asset as 'OVERVALUED' to protect capital.

Profile 20: HINDALCO INDUSTRIES LTD (HINDALCO.NS)

Sector Classification: Basic Materials

Current Market Price: ₹1,067.20

Algorithmic Intrinsic Value: ₹250.86

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Basic Materials sector, HINDALCO INDUSTRIES LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 12.49%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the

12.49% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹250.86.

Signal Conclusion: As the derived intrinsic value (₹250.86) fails to exceed the current market trading price (₹1,067.20) with a sufficient margin of safety (reflecting a premium of approximately 76.5%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 21: HINDUSTAN UNILEVER LTD. (HINDUNILVR.NS)

Sector Classification: Consumer Defensive

Current Market Price: ₹2,314.40

Algorithmic Intrinsic Value: ₹682.47

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Defensive sector, HINDUSTAN UNILEVER LTD. was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.0%. This 2.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows

(FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹682.47.

Signal Conclusion: As the derived intrinsic value (₹682.47) fails to exceed the current market trading price (₹2,314.40) with a sufficient margin of safety (reflecting a premium of approximately 70.5%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 22: ICICI BANK LTD. (ICICIBANK.NS)

Sector Classification: Financial Services

Current Market Price: ₹1,281.00

Algorithmic Intrinsic Value: ₹1,629.94

Final Investment Signal: **UNDERVALUED (BUY)**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, ICICI BANK LTD. was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the

10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,629.94.

Signal Conclusion: Because the derived intrinsic value (₹1,629.94) significantly exceeds the current market trading price (₹1,281.00), representing a calculated upside of approximately 27.2%, the pipeline formally issued an 'UNDERVALUED (BUY)' recommendation. This confirms that the asset offers a sufficient margin of safety according to the model's rigorous parameters.

Profile 23: INTERGLOBE AVIATION LTD (INDIGO.NS)

Sector Classification: Industrials

Current Market Price: ₹4,345.20

Algorithmic Intrinsic Value: ₹10,422.23

Final Investment Signal: **UNDERVALUED (BUY)**

Diagnostic Analysis & Execution Logic: Operating within the Industrials sector, INTERGLOBE AVIATION LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.18%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the

10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.18% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹10,422.23.

Signal Conclusion: Because the derived intrinsic value (₹10,422.23) significantly exceeds the current market trading price (₹4,345.20), representing a calculated upside of approximately 139.9%, the pipeline formally issued an 'UNDERVALUED (BUY)' recommendation. This confirms that the asset offers a sufficient margin of safety according to the model's rigorous parameters.

Profile 24: INFOSYS LIMITED (INFY.NS)

Sector Classification: Technology

Current Market Price: ₹1,167.50

Algorithmic Intrinsic Value: ₹16.94

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Technology sector, INFOSYS LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a

raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹16.94.

Signal Conclusion: As the derived intrinsic value (₹16.94) fails to exceed the current market trading price (₹1,167.50) with a sufficient margin of safety (reflecting a premium of approximately 98.5%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 25: ITC LTD (ITC.NS)

Sector Classification: Consumer Defensive

Current Market Price: ₹316.25

Algorithmic Intrinsic Value: ₹203.46

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Defensive sector, ITC LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure

emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 5.83%. This 5.83% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹203.46.

Signal Conclusion: As the derived intrinsic value (₹203.46) fails to exceed the current market trading price (₹316.25) with a sufficient margin of safety (reflecting a premium of approximately 35.7%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 26: JIO FIN SERVICES LTD (JIOFIN.NS)

Sector Classification: Financial Services

Current Market Price: ₹250.71

Algorithmic Intrinsic Value: ₹3.03

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, JIO FIN SERVICES LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.29%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.0%. This 2.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.29% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹3.03.

Signal Conclusion: As the derived intrinsic value (₹3.03) fails to exceed the current market trading price (₹250.71) with a sufficient margin of safety (reflecting a premium of approximately 98.8%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 27: JSW STEEL LIMITED (JSWSTEEL.NS)

Sector Classification: Basic Materials

Current Market Price: ₹1,279.70

Algorithmic Intrinsic Value: ₹498.08

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Basic Materials sector, JSW STEEL LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.47%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's

institutional friction parameters, the effective starting SGR was capped at 4.0%. This 4.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.47% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹498.08.

Signal Conclusion: As the derived intrinsic value (₹498.08) fails to exceed the current market trading price (₹1,279.70) with a sufficient margin of safety (reflecting a premium of approximately 61.1%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 28: KOTAK MAHINDRA BANK LTD (KOTAKBANK.NS)

Sector Classification: Financial Services

Current Market Price: ₹382.00

Algorithmic Intrinsic Value: ₹289.92

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, KOTAK MAHINDRA BANK LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a

raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹289.92.

Signal Conclusion: As the derived intrinsic value (₹289.92) fails to exceed the current market trading price (₹382.00) with a sufficient margin of safety (reflecting a premium of approximately 24.1%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 29: LARSEN & TOUBRO LTD. (LT.NS)

Sector Classification: Industrials

Current Market Price: ₹4,096.10

Algorithmic Intrinsic Value: ₹633.02

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Industrials sector, LARSEN & TOUBRO LTD. was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure

emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹633.02.

Signal Conclusion: As the derived intrinsic value (₹633.02) fails to exceed the current market trading price (₹4,096.10) with a sufficient margin of safety (reflecting a premium of approximately 84.5%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 30: MAHINDRA & MAHINDRA LTD (M&M.NS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹3,152.30

Algorithmic Intrinsic Value: ₹1,190.20

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, MAHINDRA & MAHINDRA LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's

specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,190.20.

Signal Conclusion: As the derived intrinsic value (₹1,190.20) fails to exceed the current market trading price (₹3,152.30) with a sufficient margin of safety (reflecting a premium of approximately 62.2%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 31: MARUTI SUZUKI INDIA LTD. (MARUTLNS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹13,257.00

Algorithmic Intrinsic Value: ₹2,055.85

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, MARUTI SUZUKI INDIA LTD. was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM)

fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 14.6%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 14.6% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹2,055.85.

Signal Conclusion: As the derived intrinsic value (₹2,055.85) fails to exceed the current market trading price (₹13,257.00) with a sufficient margin of safety (reflecting a premium of approximately 84.5%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 32: MAX HEALTHCARE INS LTD (MAXHEALTH.NS)

Sector Classification: Healthcare

Current Market Price: ₹1,006.60

Algorithmic Intrinsic Value: ₹93.18

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Healthcare sector, MAX HEALTHCARE INS LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental

disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹93.18.

Signal Conclusion: As the derived intrinsic value (₹93.18) fails to exceed the current market trading price (₹1,006.60) with a sufficient margin of safety (reflecting a premium of approximately 90.7%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 33: NESTLE INDIA LIMITED (NESTLEIND.NS)

Sector Classification: Consumer Defensive

Current Market Price: ₹1,465.60

Algorithmic Intrinsic Value: ₹88.35

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Defensive sector, NESTLE INDIA LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹88.35.

Signal Conclusion: As the derived intrinsic value (₹88.35) fails to exceed the current market trading price (₹1,465.60) with a sufficient margin of safety (reflecting a premium

of approximately 94.0%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 34: NTPC LTD (NTPC.NS)

Sector Classification: Utilities

Current Market Price: ₹401.30

Algorithmic Intrinsic Value: ₹160.01

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Utilities sector, NTPC LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 6.67%. This 6.67% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹160.01.

Signal Conclusion: As the derived intrinsic value (₹160.01) fails to exceed the current market trading price (₹401.30) with a sufficient margin of safety (reflecting a premium of

approximately 60.1%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 35: OIL AND NATURAL GAS CORP. (ONGC.NS)

Sector Classification: Energy

Current Market Price: ₹301.40

Algorithmic Intrinsic Value: ₹469.68

Final Investment Signal: UNDERVALUED (BUY)

Diagnostic Analysis & Execution Logic: Operating within the Energy sector, OIL AND NATURAL GAS CORP. was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 6.27%. This 6.27% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹469.68.

Signal Conclusion: Because the derived intrinsic value (₹469.68) significantly exceeds the current market trading price (₹301.40), representing a calculated upside of approximately 55.8%, the pipeline formally issued an 'UNDERVALUED (BUY)' recommendation. This confirms that the asset offers a sufficient margin of safety according to the model's rigorous parameters.

Profile 36: POWER GRID CORP. LTD. (POWERGRID.NS)

Sector Classification: Utilities

Current Market Price: ₹320.35

Algorithmic Intrinsic Value: ₹222.52

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Utilities sector, POWER GRID CORP. LTD. was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 7.08%. This 7.08% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the

11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹222.52.

Signal Conclusion: As the derived intrinsic value (₹222.52) fails to exceed the current market trading price (₹320.35) with a sufficient margin of safety (reflecting a premium of approximately 30.5%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 37: RELIANCE INDUSTRIES LTD (RELIANCE.NS)

Sector Classification: Energy

Current Market Price: ₹1,425.40

Algorithmic Intrinsic Value: ₹897.55

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Energy sector, RELIANCE INDUSTRIES LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 8.11%. This 8.11% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows

(FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹897.55.

Signal Conclusion: As the derived intrinsic value (₹897.55) fails to exceed the current market trading price (₹1,425.40) with a sufficient margin of safety (reflecting a premium of approximately 37.0%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 38: SBI LIFE INSURANCE CO LTD (SBILIFE.NS)

Sector Classification: Financial Services

Current Market Price: ₹1,816.20

Algorithmic Intrinsic Value: ₹0.00

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, SBI LIFE INSURANCE CO LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the

10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹0.00.

Signal Conclusion: The algorithm determined an intrinsic value of ₹0.00. This severe downward adjustment typically occurs when extreme, sustained capital expenditure or structural financial anomalies prevent the normalization of positive cash flows, leading the algorithm to defensively classify the asset as 'OVERVALUED' to protect capital.

Profile 39: SHRIRAM FINANCE LIMITED (SHRIRAMFIN.NS)

Sector Classification: Financial Services

Current Market Price: ₹956.85

Algorithmic Intrinsic Value: ₹94.30

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, SHRIRAM FINANCE LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.51%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This

10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.51% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹94.30.

Signal Conclusion: As the derived intrinsic value (₹94.30) fails to exceed the current market trading price (₹956.85) with a sufficient margin of safety (reflecting a premium of approximately 90.1%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 40: STATE BANK OF INDIA (SBIN.NS)

Sector Classification: Financial Services

Current Market Price: ₹1,086.90

Algorithmic Intrinsic Value: ₹852.47

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Financial Services sector, STATE BANK OF INDIA was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's

institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹852.47.

Signal Conclusion: As the derived intrinsic value (₹852.47) fails to exceed the current market trading price (₹1,086.90) with a sufficient margin of safety (reflecting a premium of approximately 21.6%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 41: SUN PHARMACEUTICAL IND L (SUNPHARMA.NS)

Sector Classification: Healthcare

Current Market Price: ₹1,778.70

Algorithmic Intrinsic Value: ₹910.06

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Healthcare sector, SUN PHARMACEUTICAL IND L was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 9.81%. This 9.81% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹910.06.

Signal Conclusion: As the derived intrinsic value (₹910.06) fails to exceed the current market trading price (₹1,778.70) with a sufficient margin of safety (reflecting a premium of approximately 48.8%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 42: TATA CONSULTANCY SERV LT (TCS.NS)

Sector Classification: Technology

Current Market Price: ₹2,474.70

Algorithmic Intrinsic Value: ₹2,431.59

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Technology sector, TATA CONSULTANCY SERV LT was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure

emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹2,431.59.

Signal Conclusion: As the derived intrinsic value (₹2,431.59) fails to exceed the current market trading price (₹2,474.70) with a sufficient margin of safety (reflecting a premium of approximately 1.7%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 43: TATA CONSUMER PRODUCT LTD (TATACONSUM.NS)

Sector Classification: Consumer Defensive

Current Market Price: ₹1,168.00

Algorithmic Intrinsic Value: ₹249.03

Final Investment Signal: OVERVALUED

Diagnostic Analysis & Execution Logic: Operating within the Consumer Defensive sector, TATA CONSUMER PRODUCT LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's

specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.84%. This 2.84% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹249.03.

Signal Conclusion: As the derived intrinsic value (₹249.03) fails to exceed the current market trading price (₹1,168.00) with a sufficient margin of safety (reflecting a premium of approximately 78.7%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 44: TATA MOTORS PASS VEH LTD (TMPV.NS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹352.70

Algorithmic Intrinsic Value: ₹1,072.36

Final Investment Signal: **UNDERVALUED (BUY)**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, TATA MOTORS PASS VEH LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035

benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 12.06%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 12.06% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹1,072.36.

Signal Conclusion: Because the derived intrinsic value (₹1,072.36) significantly exceeds the current market trading price (₹352.70), representing a calculated upside of approximately 204.0%, the pipeline formally issued an 'UNDERVALUED (BUY)' recommendation. This confirms that the asset offers a sufficient margin of safety according to the model's rigorous parameters.

Profile 45: TATA STEEL LIMITED (TATASTEEL.NS)

Sector Classification: Basic Materials

Current Market Price: ₹215.88

Algorithmic Intrinsic Value: ₹73.72

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Basic Materials sector, TATA STEEL LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the

Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 12.89%.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.0%. This 2.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 12.89% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹73.72.

Signal Conclusion: As the derived intrinsic value (₹73.72) fails to exceed the current market trading price (₹215.88) with a sufficient margin of safety (reflecting a premium of approximately 65.9%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 46: TECH MAHINDRA LIMITED (TECHM.NS)

Sector Classification: Technology

Current Market Price: ₹1,459.80

Algorithmic Intrinsic Value: ₹952.20

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Technology sector, TECH MAHINDRA LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold;

therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.76%. This 2.76% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹952.20.

Signal Conclusion: As the derived intrinsic value (₹952.20) fails to exceed the current market trading price (₹1,459.80) with a sufficient margin of safety (reflecting a premium of approximately 34.8%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 47: TITAN COMPANY LIMITED (TITAN.NS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹4,439.80

Algorithmic Intrinsic Value: ₹624.16

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, TITAN COMPANY LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of

11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹624.16.

Signal Conclusion: As the derived intrinsic value (₹624.16) fails to exceed the current market trading price (₹4,439.80) with a sufficient margin of safety (reflecting a premium of approximately 85.9%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 48: TRENT LTD (TRENT.NS)

Sector Classification: Consumer Cyclical

Current Market Price: ₹4,228.30

Algorithmic Intrinsic Value: ₹467.40

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Consumer Cyclical sector, TRENT LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-

Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 10.0%. This indicates the firm's historical profitability metrics were exceptionally high, triggering the 10% maximum growth cap to prevent the 'Gordon Growth Explosion' anomaly. This 10.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹467.40.

Signal Conclusion: As the derived intrinsic value (₹467.40) fails to exceed the current market trading price (₹4,228.30) with a sufficient margin of safety (reflecting a premium of approximately 88.9%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 49: ULTRATECH CEMENT LIMITED (ULTRACEMCO.NS)

Sector Classification: Basic Materials

Current Market Price: ₹11,833.00

Algorithmic Intrinsic Value: ₹3,313.76

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Basic Materials sector, ULTRATECH CEMENT LIMITED was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 7.49%. This 7.49% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹3,313.76.

Signal Conclusion: As the derived intrinsic value (₹3,313.76) fails to exceed the current market trading price (₹11,833.00) with a sufficient margin of safety (reflecting a premium of approximately 72.0%), the pipeline strictly classified the equity as 'OVERVALUED'. This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

Profile 50: WIPRO LTD (WIPRO.NS)

Sector Classification: Technology

Current Market Price: ₹200.68

Algorithmic Intrinsic Value: ₹222.69

Final Investment Signal: **OVERVALUED**

Diagnostic Analysis & Execution Logic: Operating within the Technology sector, WIPRO LTD was processed through the automated ingestion pipeline. The system successfully retrieved its most recent trailing twelve-month (TTM) fundamental disclosures via the yfinance API. Concurrently, the macro-engine scraped the live Risk-Free Rate from the Reserve Bank of India (RBI) GS 2035 benchmark. Utilizing the Capital Asset Pricing Model (CAPM), adjusting for the asset's specific Beta, the algorithm computed a dynamic Weighted Average Cost of Capital (WACC) of 11.0%. Notably, the computed CAPM rate was lower than the institutional risk threshold; therefore, the pipeline actively enforced the programmed 11.0% WACC floor to ensure emerging market risks and currency fluctuations were adequately priced into the discount rate.

Growth Trajectory & Discounting: The programmatic 3-step DuPont analysis evaluated the firm's net profit margin, asset turnover, and equity multiplier to derive a raw Sustainable Growth Rate (SGR). Following the application of the model's institutional friction parameters, the effective starting SGR was capped at 2.0%. This 2.0% metric was subsequently subjected to the pipeline's mean-reverting decay function over a 5-year explicit projection period. The projected normalized Free Cash Flows (FCFF), alongside the terminal value, were discounted back to present value using the 11.0% rate. Dividing by total shares outstanding, the algorithm determined an absolute intrinsic value per share of ₹222.69.

Signal Conclusion: As the derived intrinsic value (₹222.69) fails to exceed the current market trading price (₹200.68) with a sufficient margin of safety (reflecting a premium of approximately -11.0%), the pipeline strictly classified the equity as 'OVERVALUED'.

This programmatic outcome suggests the asset's current market valuation is driven by speculative premiums rather than underlying cash flow realities.

CHAPTER 5: CONCLUSION & IMPLICATIONS

5.1 Summary of Findings

The primary objective of this dissertation was to bridge the structural gap between traditional corporate finance theory and modern data science engineering. By successfully architecting, coding, and deploying an end-to-end automated valuation pipeline in Python, the research proved that full-stack programmatic architectures can fundamentally optimize the equity valuation workflow.

The multi-phase pipeline effortlessly ingested live market data via APIs, persisted it securely in a relational SQLite database, and executed rigorous financial engineering. Crucially, the implementation of institutional safeguards—such as cash flow smoothing, mean-reverting growth decay functions, and a hard 11% WACC floor—successfully mitigated the 'algorithmic over-optimism' that severely plagues standard automated models. Consequently, the pipeline generated highly realistic, defensible investment signals, categorizing only 10% of the Nifty 50 index as undervalued bargains in the current premium market environment.

5.2 Conclusion

In conclusion, the transition to dynamic, database-backed programmatic pipelines is not merely a technological novelty for financial analysts; it is a necessary evolution for modern quantitative analysis. By removing static heuristic biases and forcing mathematical models to rely on live macroeconomic benchmarks scraped directly from central banking institutions (like the RBI), portfolio managers can produce valuations that are both infinitely scalable and mathematically defensible. The success of this codebase proves that complex financial theory can be reliably translated into autonomous software execution.

5.3 Strategic Implications

The findings and the resulting codebase of this study carry significant strategic implications for the field of finance:

1. **Full-Stack Capabilities Required:** The successful integration of SQL, Python, and the Streamlit framework demonstrates that modern business students and financial analysts must evolve beyond spreadsheet mechanics. Familiarity with code is becoming a prerequisite for advanced financial modeling.
2. **Bias Reduction through Code:** Programmatic execution removes emotional sentiment and confirmation bias from valuation. By forcing the algorithm to strictly adhere to the defined WACC floors and growth caps, investment decisions are forced to align strictly with cash flow realities rather than market hype.
3. **Real-Time Deployment and Visualization:** The successful creation of the interactive Streamlit dashboard proves that complex quantitative outputs—typically buried in the backend—can be made instantly accessible and visually digestible for executive decision-makers through modern web frameworks.

5.4 Recommendations for Future Research

While the architecture developed in this study is production-ready and highly robust, it serves as a foundational framework for further expansion. Future research and development could focus on:

- **Cloud Database Migration:** Transitioning the local SQLite database to a cloud-hosted PostgreSQL server (e.g., AWS RDS) to allow for distributed team access and continuous serverless execution via cron jobs.
- **Machine Learning Time-Series Forecasting:** Expanding the valuation logic to include Long Short-Term Memory (LSTM) neural networks to forecast future revenue growth rates based on decades of historical data, rather than relying solely on the DuPont SGR formula.
- **Relative Valuation Integration:** Programming the pipeline to simultaneously scrape and calculate relative valuation multiples (such as dynamic P/E and EV/EBITDA sector

averages) alongside the intrinsic DCF model to provide a more holistic algorithmic consensus signal.

REFERENCES

- Chandra, P. (2017). Investment analysis and portfolio management (5th ed.). McGraw-Hill Education.
- Damodaran, A. (2012). Investment valuation: Tools and techniques for determining the value of any asset (3rd ed.). John Wiley & Sons.
- Dixit, A., & Kumar, S. (2020). Algorithmic trading and fundamental analysis in the Indian equity market. *Journal of Emerging Market Finance*, 19(2), 145-168.
- Hull, J. C. (2018). Options, futures, and other derivatives (10th ed.). Pearson.
- Koller, T., Goedhart, M., & Wessels, D. (2020). Valuation: Measuring and managing the value of companies (7th ed.). McKinsey & Company. John Wiley & Sons.
- McKinney, W. (2012). Python for data analysis: Data wrangling with Pandas, NumPy, and IPython. O'Reilly Media.

APPENDICES

Appendix A: Algorithmic Pipeline Source Code (engine.py)

The following section contains the complete, unedited Python source code for the automated data ingestion, macroeconomic engine, and defensive valuation architecture. The script incorporates web scraping modules, `yfinance` API calls, SQLite database commands, and the algorithmic DCF math functions with institutional friction protocols.

```
import pandas as pd
import yfinance as yf
import sqlite3
import time
import requests
import numpy as np
import datetime
import re
import warnings
from io import StringIO
from bs4 import BeautifulSoup

# Suppress warnings for clean console output
warnings.filterwarnings('ignore')

# =====
# MODULE 1: DYNAMIC MACROECONOMIC ENGINE
# =====
def get_dynamic_macro_metrics():
    print("Initializing Dynamic Macroeconomic Engine...")
    rf_rate = None

    # 1. Targeted RBI Scraper (10-Year Benchmark GS 2035)
    print(" [*] Attempting to fetch 10-Year Yield (GS 2035) from RBI...")
    try:
        url = "https://www.rbi.org.in/"
        headers = {"User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64)"}
        response = requests.get(url, headers=headers, timeout=15)
        soup = BeautifulSoup(response.text, 'html.parser')

        strings = list(soup.stripped_strings)
        for i, text in enumerate(strings):
            if "GS 2035" in text:
                potential_val = strings[i+1]
                match = re.search(r'([0-9]+\.[0-9]+)', potential_val)
```



```

        if match:
            rf_rate = float(match.group(1)) / 100
            print(f" [+] SUCCESS: Live 10-Yr Risk-Free Rate ({rf_rate*100:.4f}%)
fetched via RBI.")
            break
    except Exception as e:
        print(f" [-] RBI Scraper failed: {e}")

    if rf_rate is None:
        print(" [!] Could not locate GS 2035. Defaulting to 7.05% baseline.")
        rf_rate = 0.0705

# 2. Market Return (Nifty 50 10-Yr CAGR)
try:
    nifty = yf.Ticker("^NSEI")
    nifty_hist = nifty.history(period="10y")
    start, end = nifty_hist['Close'].iloc[0], nifty_hist['Close'].iloc[-1]
    market_return = (end / start) ** (1 / 10) - 1
    print(f" [+] Live Nifty 50 10-Yr CAGR ({market_return*100:.2f}%) fetched.")
except:
    market_return = 0.12 # Standard baseline

erp = market_return - rf_rate
terminal_growth = min(rf_rate * 0.70, 0.045) # Conservative Cap

return {'Risk_Free_Rate': rf_rate, 'Equity_Risk_Premium': erp,
'Terminal_Growth_Rate': terminal_growth}

# =====
# MODULE 2: DATABASE & INGESTION
# =====
def setup_database():
    conn = sqlite3.connect('nifty_finance.db')
    cursor = conn.cursor()
    cursor.execute("""CREATE TABLE IF NOT EXISTS fundamental_data (
        ticker TEXT PRIMARY KEY, company_name TEXT, sector TEXT, net_income
REAL,
        total_revenue REAL, total_assets REAL, total_equity REAL, operating_cash_flow
REAL,
        capital_expenditure REAL, shares_outstanding REAL, current_price REAL,
        last_updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP)""")
    conn.commit()
    return conn

def get_nifty50_tickers():
    print("Scraping live Nifty 50 constituents...")

```

```

url = 'https://en.wikipedia.org/wiki/NIFTY_50'
headers = {'User-Agent': 'Mozilla/5.0'}
response = requests.get(url, headers=headers)
tables = pd.read_html(StringIO(response.text))
for table in tables:
    if 'Symbol' in table.columns:
        return [symbol + '.NS' for symbol in table['Symbol'].tolist()]

def extract_and_store(conn, tickers):
    cursor = conn.cursor()
    print("Initiating API Extraction Pipeline...")
    for ticker in tickers:
        print(f" -> Processing {ticker}...")
        try:
            stock = yf.Ticker(ticker)
            info, inc, bs, cf = stock.info, stock.financials, stock.balance_sheet, stock.cashflow
            def get_val(df, key): return df.loc[key].iloc[0] if key in df.index else 0

            equity = get_val(bs, 'Stockholders Equity') or get_val(bs, 'Total Equity Gross
Minority Interest')

            cursor.execute("INSERT OR REPLACE INTO fundamental_data
(ticker, company_name, sector, net_income, total_revenue, total_assets,
total_equity, operating_cash_flow, capital_expenditure, shares_outstanding,
current_price)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)", (
            ticker, info.get('shortName', ticker), info.get('sector', 'Unknown'),
            get_val(inc, 'Net Income'), get_val(inc, 'Total Revenue'),
            get_val(bs, 'Total Assets'), equity,
            get_val(cf, 'Operating Cash Flow'), get_val(cf, 'Capital Expenditure'),
            info.get('sharesOutstanding', 1), info.get('currentPrice', 0.0)
        ))
            conn.commit()
            time.sleep(0.5)
        except Exception: pass

# =====
# MODULE 3: INSTITUTIONAL VALUATION ENGINE
# =====

def execute_pipeline(conn, macro):
    print("Executing DCF Engine with Cash Flow Smoothing & Growth Decay...")
    df = pd.read_sql_query("SELECT * FROM fundamental_data", conn)
    df.replace(0, np.nan, inplace=True)

    # 1. Engineering DuPont ROE
    df['net_profit_margin'] = df['net_income'] / df['total_revenue']

```

```

df['asset_turnover'] = df['total_revenue'] / df['total_assets']
df['equity_multiplier'] = df['total_assets'] / df['total_equity']
df['programmatic_roe'] = df['net_profit_margin'] * df['asset_turnover'] *
df['equity_multiplier']

# 2. Institutional Cash Flow Smoothing
# Assume a maintenance FCFF of 5% of Revenue if current FCFF is negative/distorted
df['raw_fcff'] = df['operating_cash_flow'] + df['capital_expenditure']
df['normalized_fcff'] = np.where(df['raw_fcff'] <= 0, df['total_revenue'] * 0.05,
df['raw_fcff'])

df.fillna(0, inplace=True)

intrinsic_values, signals, dynamic_waccs, dynamic_sgrs = [], [], [], []

for row in df.itertuples():
    try:
        stock_info = yf.Ticker(row.ticker).info
        beta = stock_info.get('beta', 1.1) or 1.1
        payout = stock_info.get('payoutRatio', 0.0) or 0.2

        # --- SAFEGUARD: WACC Floor (Institutional Practice for Emerging Markets) -
--
        wacc = macro['Risk_Free_Rate'] + (beta * (macro['Equity_Risk_Premium'] +
0.02)) # +2% Country Risk
        wacc = max(wacc, 0.11) # 11% Floor

        # --- SAFEGUARD: Growth Cap ---
        initial_g = min(max(row.programmatic_roe * (1 - payout), 0.02), 0.10) # 10%
Cap

        dynamic_waccs.append(wacc); dynamic_sgrs.append(initial_g)

        # --- SAFEGUARD: Mean-Reverting Growth (Fade growth toward terminal rate)
---
        p_fcff = []
        temp_fcff = row.normalized_fcff
        temp_g = initial_g
        for i in range(1, 6):
            temp_fcff *= (1 + temp_g)
            p_fcff.append(temp_fcff)
            temp_g = (temp_g * 0.7) + (macro['Terminal_Growth_Rate'] * 0.3)

        pv_explicit = sum([cf / (1 + wacc)**i for i, cf in enumerate(p_fcff, 1)])

        # Terminal Value

```

```

term_g = min(macro['Terminal_Growth_Rate'], wacc - 0.05)
terminal_val = (p_ff[-1] * (1 + term_g)) / (wacc - term_g)
intrinsic = (pv_explicit + (terminal_val / (1 + wacc)**5)) /
row.shares_outstanding

intrinsic_values.append(np.round(intrinsic, 2))
# Margin of Safety Signal (Only BUY if 20% upside exists)
signals.append("UNDERVALUED (BUY)" if intrinsic > (row.current_price *
1.2) else "OVERVALUED")

except Exception:
    intrinsic_values.append(0); signals.append("ERROR");
dynamic_waccs.append(0); dynamic_sgrs.append(0)

df['dynamic_wacc'], df['dynamic_sgr'] = dynamic_waccs, dynamic_sgrs
df['intrinsic_value'], df['signal'] = intrinsic_values, signals
df.to_sql('final_valuations', conn, if_exists='replace', index=False)

if __name__ == "__main__":
    db_conn = setup_database()
    macros = get_dynamic_macro_metrics()
    tickers = get_nifty50_tickers()
    extract_and_store(db_conn, tickers)
    execute_pipeline(db_conn, macros)
    db_conn.close()
    print("\n--- Pipeline Complete. Run 'streamlit run app.py' ---")

```

Appendix B: Interactive Dashboard Source Code (app.py)

The following section contains the complete Python source code utilizing the Streamlit framework. This script establishes a connection to the local database, queries the final output table, and renders the interactive web dashboard for visual analysis.

```
import streamlit as st
import sqlite3
import pandas as pd

st.set_page_config(page_title="Institutional Valuation Dashboard", layout="wide")
st.title("📊 Nifty 50 Algorithmic Valuation Dashboard")
st.markdown("Automated DCF Engine with **Growth Decay** and **WACC Spread Safeguards**.")

@st.cache_data
def load_data():
    conn = sqlite3.connect('nifty_finance.db')
    master_df = pd.read_sql_query("SELECT * FROM final_valuations", conn)
    conn.close()
    return master_df

df = load_data()

st.sidebar.header("⚙️ Dashboard Filters")
selected_signal = st.sidebar.multiselect("Investment Signal",
options=df['signal'].unique(), default=df['signal'].unique())
selected_sector = st.sidebar.multiselect("Sector Filter", options=df['sector'].unique(),
default=df['sector'].unique())

filtered_df = df[(df['signal'].isin(selected_signal)) & (df['sector'].isin(selected_sector))]

st.markdown("#### Pipeline Analytics")
col1, col2, col3 = st.columns(3)
col1.metric("Total Equities", len(filtered_df))
col2.metric("Conservative Buy Signals", len(filtered_df[filtered_df['signal'] ==
'UNDERVALUED (BUY)']))
col3.metric("Negative Cash Flow", len(filtered_df[filtered_df['signal'] == 'NEGATIVE
CASH FLOW']))

st.divider()
st.subheader("Defensive Valuation Output")
display_cols = ['ticker', 'company_name', 'sector', 'current_price', 'intrinsic_value', 'signal',
'dynamic_wacc', 'dynamic_sgr']
```

```

styled_df = filtered_df[display_cols].copy()
styled_df['dynamic_wacc'] = (styled_df['dynamic_wacc'] * 100).round(2).astype(str) +
'%'
styled_df['dynamic_sgr'] = (styled_df['dynamic_sgr'] * 100).round(2).astype(str) + '%'

st.dataframe(styled_df, use_container_width=True)

st.subheader("Valuation Gap Analysis")
chart_df = filtered_df[filtered_df['intrinsic_value'] > 0]
if not chart_df.empty:
    st.bar_chart(chart_df.set_index('ticker')[['current_price', 'intrinsic_value']])

```

Appendix C: Web Application Interface

